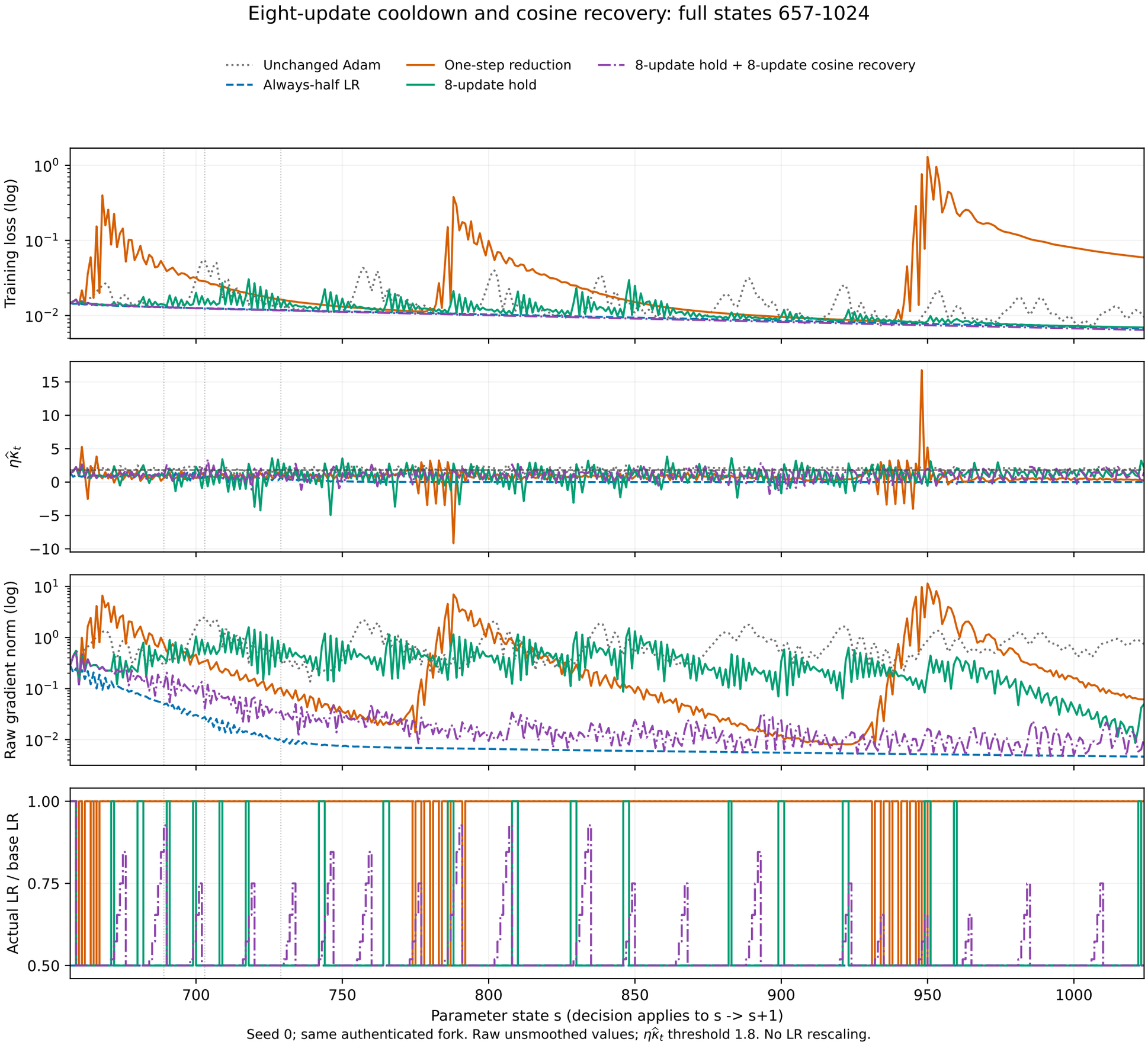

Eight-update cooldown and cosine recovery: full states 657-1024
Unchanged Adam
Always-half LR
One-step reduction
8-update hold
8-update hold + 8-update cosine recovery
Training loss (log)
$\eta\hat{\kappa}_t$
Raw gradient norm (log)
Actual LR / base LR
Parameter state s (decision applies to s -> s+1)
Seed 0; same authenticated fork. Raw unsmoothed values; $\eta\hat{\kappa}_t$ threshold 1.8. No LR rescaling.